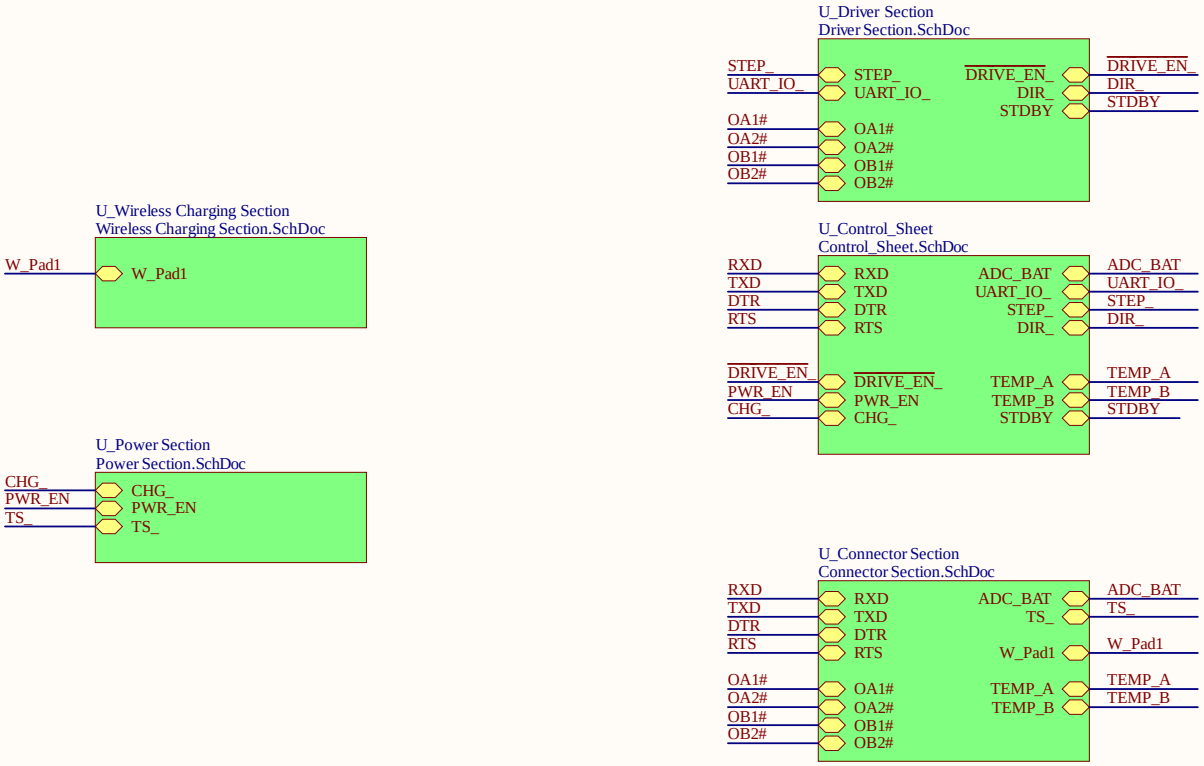
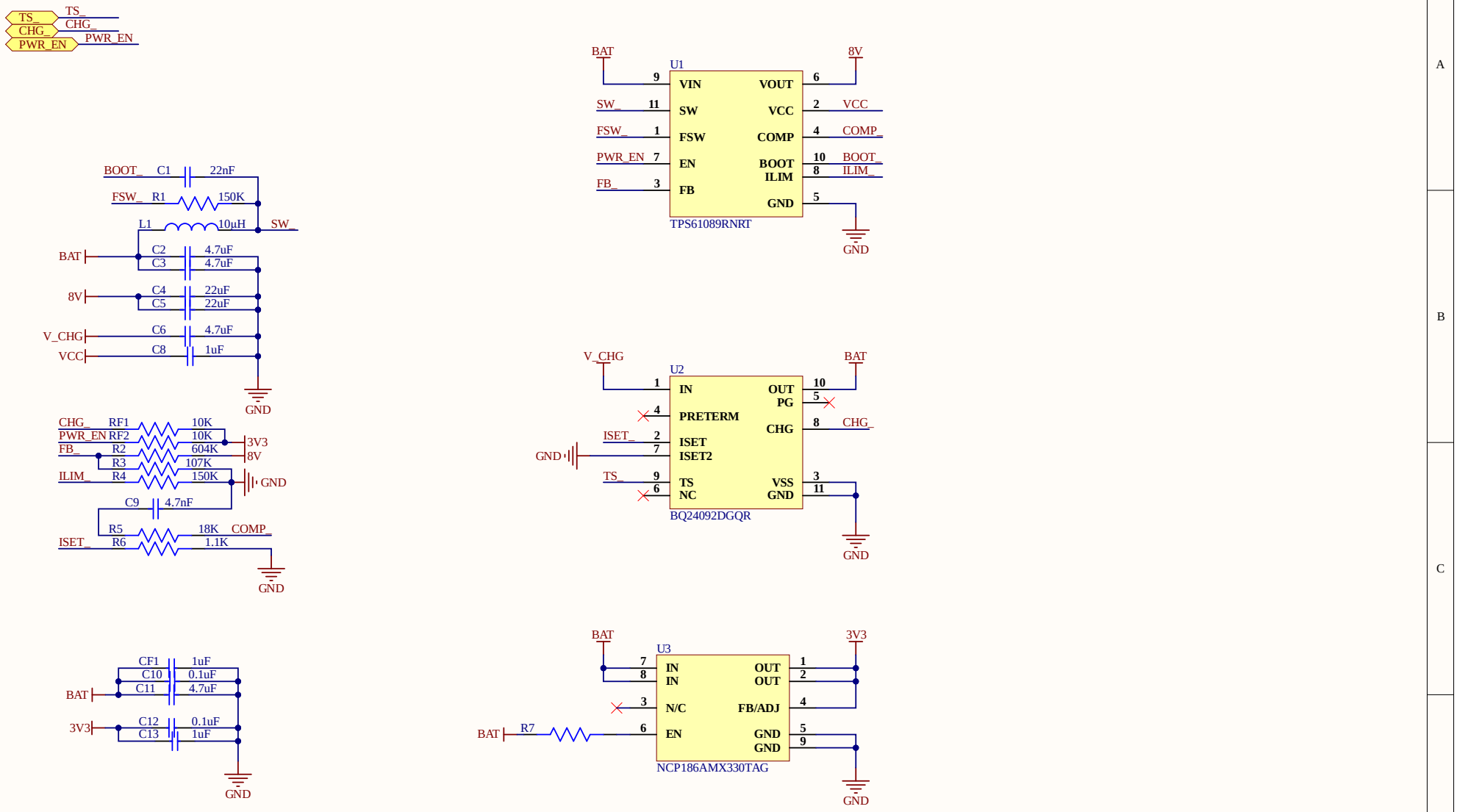
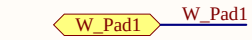
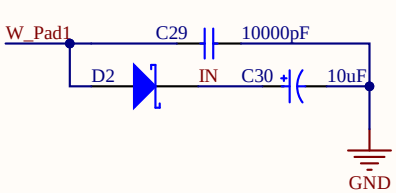
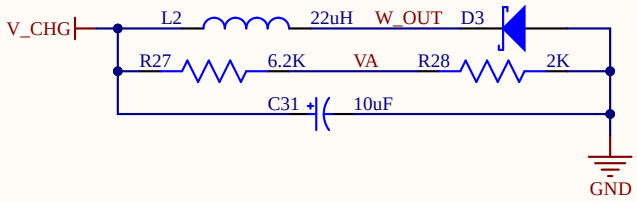

U_Driver Section
Driver Section.SchDoc
STEP_
UART_IO_
OA1#
OA2#
OB1#
OB2#
STEP_
UART_IO_
OA1#
OA2#
OB1#
OB2#
DRIVE_EN_
DIR_
STDBY
DRIVE_EN_
DIR_
STDBY
U_Wireless Charging Section
Wireless Charging Section.SchDoc
W_Pad1
W_Pad1
U_Control_Sheet
Control_Sheet.SchDoc
RXD
TXD
DTR
RTS
DRIVE_EN_
PWR_EN
CHG_
RXD
TXD
DTR
RTS
DRIVE_EN_
PWR_EN
CHG_
ADC_BAT
UART_IO_
STEP_
DIR_
TEMP_A
TEMP_B
STDBY
ADC_BAT
UART_IO_
STEP_
DIR_
TEMP_A
TEMP_B
STDBY
U_Power Section
Power Section.SchDoc
CHG_
PWR_EN
TS_
CHG_
PWR_EN
TS_
U_Connector Section
Connector Section.SchDoc
RXD
TXD
DTR
RTS
OA1#
OA2#
OB1#
OB2#
RXD
TXD
DTR
RTS
OA1#
OA2#
OB1#
OB2#
ADC_BAT
TS_
W_Pad1
TEMP_A
TEMP_B
ADC_BAT
TS_
W_Pad1
TEMP_A
TEMP_B

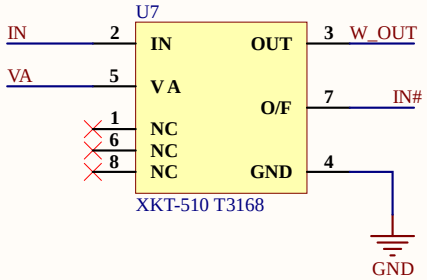

TS_
CHG_
PWR_EN
BOOT_ C1 22nF
FSW_ R1 150K
L1 10µH SW_
BAT
C2 4.7uF
C3 4.7uF
8V
C4 22uF
C5 22uF
V_CHG
C6 4.7uF
VCC
C8 1uF
GND
CHG_ RF1 10K
PWR_EN RF2 10K
FB_ R2 604K
R3 107K
ILIM_ R4 150K
3V3
8V
GND
C9 4.7nF
R5 18K COMP_
ISET_ R6 1.1K
GND
CF1 1uF
C10 0.1uF
BAT
C11 4.7uF
C12 0.1uF
3V3
C13 1uF
GND
BAT
U1
9 VIN
SW_ 11 SW
FSW_ 1 FSW
PWR_EN 7 EN
FB_ 3 FB
GND
VOUT 6
8V
VCC 2
VCC
COMP 4
COMP_
BOOT 10
BOOT_
ILIM 8
ILIM_
GND 5
GND
TPS61089RNRT
V_CHG
U2
1 IN
4 PRETERM
ISET_ 2 ISET
GND 7 ISET2
TS_ 9 TS
6 NC
OUT 10
PG 5
CHG 8
CHG_
VSS 3
GND 11
BAT
GND
BQ24092DGQR
BAT
U3
7 IN
8 IN
3 N/C
BAT R7 6 EN
OUT 1
OUT 2
FB/ADJ 4
GND 5
GND 9
3V3
GND
NCP186AMX330TAG

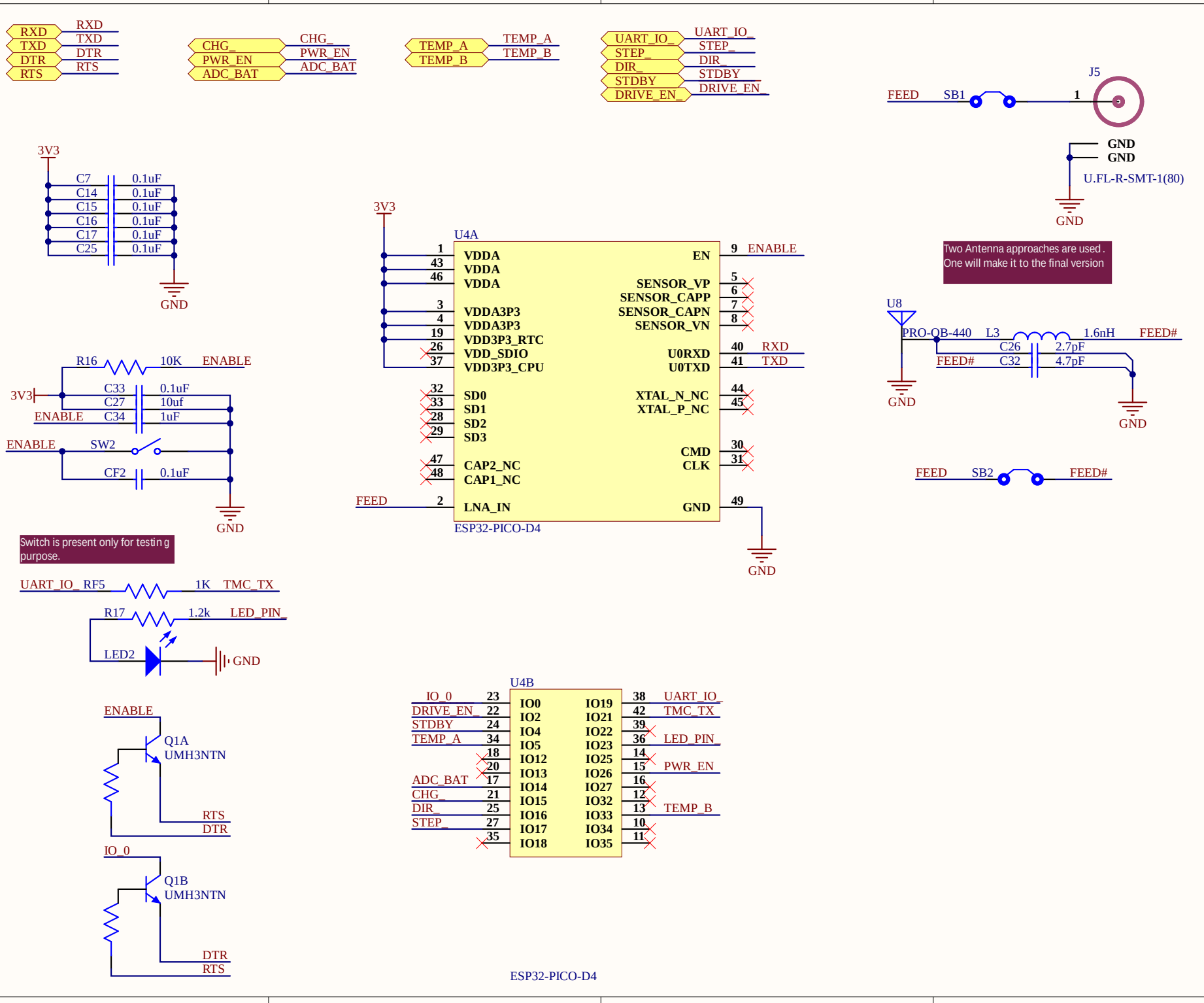

W_Pad1
W_Pad1

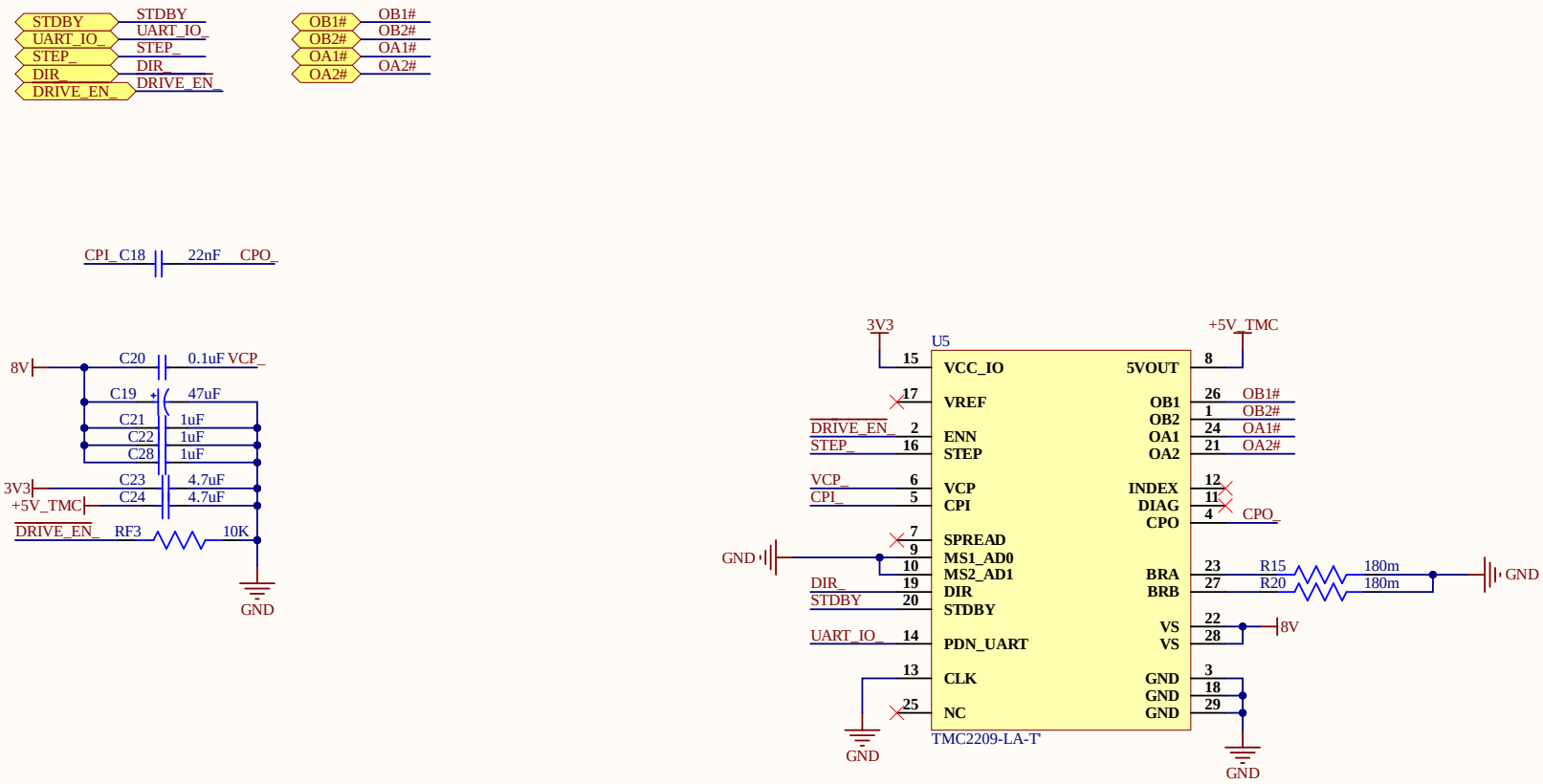

W_Pad1
C29
10000pF
D2
IN
C30
10uF
GND

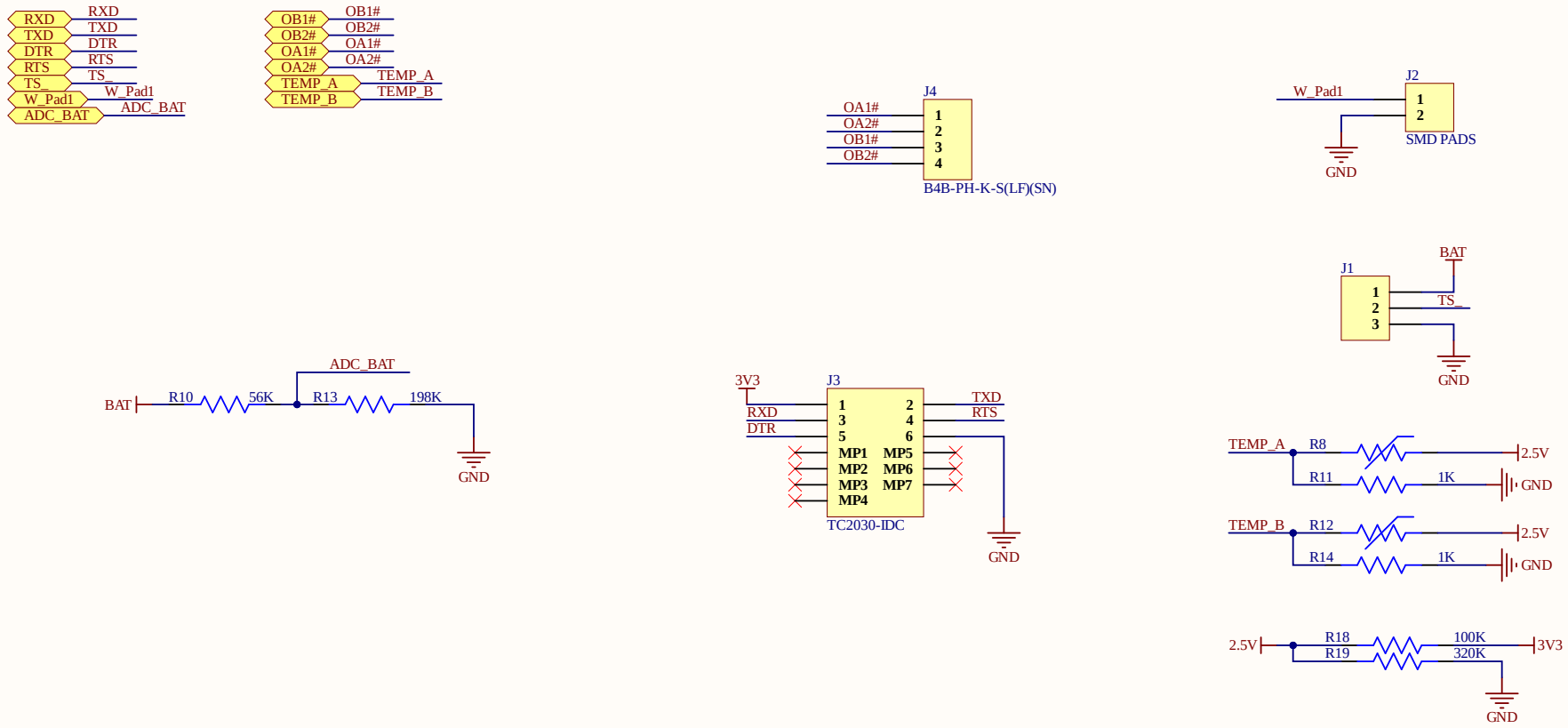

IN
R9
100K
IN#

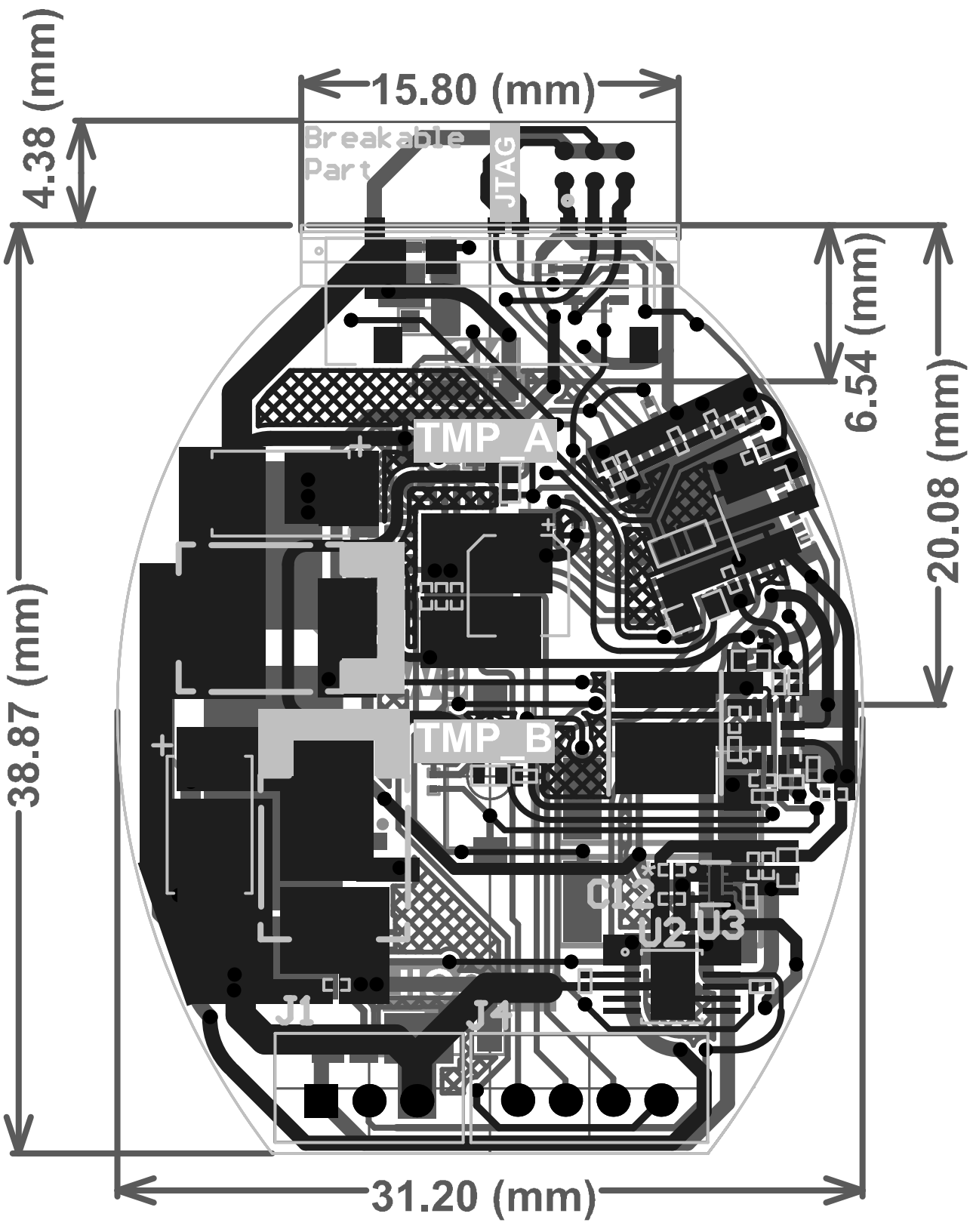

V_CHG
L2
22uH
W_OUT
D3
R27
6.2K
VA
R28
2K
C31
10uF
GND


U7
IN
2
IN
OUT
3
W_OUT
VA
5
V A
O/F
7
IN#
1
NC
6
NC
8
NC
GND
4
XKT-510 T3168
GND


RXD
TXD
DTR
RTS
CHG_
PWR_EN
ADC_BAT
TEMP_A
TEMP_B
UART_IO_
STEP_
DIR_
STDBY
DRIVE_EN_
J5
FEED
SB1
1
GND
GND
U.FL-R-SMT-1(80)
3V3
C7 0.1uF
C14 0.1uF
C15 0.1uF
C16 0.1uF
C17 0.1uF
C25 0.1uF
GND
U4A
VDDA
VDDA
VDDA
VDDA3P3
VDDA3P3
VDD3P3_RTC
VDD_SDIO
VDD3P3_CPU
SD0
SD1
SD2
SD3
CAP2_NC
CAP1_NC
LNA_IN
EN
SENSOR_VP
SENSOR_CAPP
SENSOR_CAPN
SENSOR_VN
U0RXD
U0TXD
XTAL_N_NC
XTAL_P_NC
CMD
CLK
GND
ESP32-PICO-D4
Two Antenna approaches are used .
One will make it to the final version
U8
PRO-QB-440
L3 1.6nH FEED#
C26 2.7pF
FEED# C32 4.7pF
GND
GND
FEED
SB2
FEED#
R16 10K ENABLE
3V3
C33 0.1uF
C27 10uf
ENABLE C34 1uF
ENABLE SW2
CF2 0.1uF
GND
Switch is present only for testing purpose.
UART_IO_ RF5 1K TMC_TX
R17 1.2k LED_PIN_
LED2
GND
U4B
IO0
IO2
IO4
IO5
IO12
IO13
IO14
IO15
IO16
IO17
IO18
IO19
IO21
IO22
IO23
IO25
IO26
IO27
IO32
IO33
IO34
IO35
ENABLE
Q1A
UMH3NTN
RTS
DTR
IO_0
Q1B
UMH3NTN
DTR
RTS
ESP32-PICO-D4


STDBY
UART_IO_
STEP_
DIR_
DRIVE_EN_
OB1#
OB2#
OA1#
OA2#
CPI_ C18 22nF CPO_
8V
C20 0.1uF VCP_
C19 47uF
C21 1uF
C22 1uF
C28 1uF
3V3
C23 4.7uF
+5V_TMC
C24 4.7uF
DRIVE_EN_ RF3 10K
GND
U5
3V3
+5V_TMC
15 VCC_IO
5VOUT 8
17 VREF
OB1 26 OB1#
OB2 1 OB2#
DRIVE_EN_ 2 ENN
OA1 24 OA1#
STEP_ 16 STEP
OA2 21 OA2#
VCP_ 6 VCP
INDEX 12
CPI_ 5 CPI
DIAG 11
CPO 4 CPO_
7 SPREAD
GND 9 MS1_AD0
10 MS2_AD1
BRA 23 R15 180m
DIR_ 19 DIR
BRB 27 R20 180m
GND
STDBY 20 STDBY
VS 22 8V
UART_IO_ 14 PDN_UART
VS 28
13 CLK
GND 3
GND 18
25 NC
GND 29
TMC2209-LA-T
GND
GND


RXD
TXD
DTR
RTS
TS_
W_Pad1
ADC_BAT
OB1#
OB2#
OA1#
OA2#
TEMP_A
TEMP_B
J4
B4B-PH-K-S(LF)(SN)
J2
SMD PADS
GND
J1
BAT
R10
56K
R13
198K
J3
3V3
TC2030-IDC
MP1
MP2
MP3
MP4
MP5
MP6
MP7
R8
R11
1K
2.5V
R12
R14
R18
R19
100K
320K


4.38 (mm)
15.80 (mm)
Breakable
Part
JTAG
6.54 (mm)
20.08 (mm)
38.87 (mm)
TMP_A
TMP_B
U2
U3
J1
J4
31.20 (mm)